The Prospects of Underwater Solar Energy for Sustainable Power Generation in Bangladesh

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Abstract

Bangladesh's rivers, ponds, reservoirs and coastlines offer new opportunities for renewable energy. Underwater solar uses specially designed photovoltaic devices to harvest light beneath the water's surface, with harvesting shown at around 10 m depth, to power submerged sensors, monitoring and communication systems. This proposal explores its future potential for Bangladesh, integrated with solar, wind, battery storage and intelligent energy management, rather than claiming rapid large-scale deployment.

Problem Statement

1. Solar installations on the ground are constrained by the availability of land and competing land uses.
2. Water bodies and coastal regions continue to be insufficiently utilized for renewable energy.
3. Floating PV operates on the surface; submerged PV is relatively less investigated.
4. Dim lighting, signal loss, biological growth, rust, pressure, and electrical safety pose challenges for underwater PV.
5. An evaluation specific to Bangladesh is necessary prior to execution.

Objectives

1. Assess the viability of underwater photovoltaic energy collection in Bangladesh
2. Develop a conceptual hybrid model that integrates advanced energy management, battery storage and underwater solar panels
3. Assess possible research obstacles and future directions for inland-water and coastal applications

Expected Outcomes & Impact

1. A feasibility framework for underwater PV focused on Bangladesh.
2. Research findings for future underwater PV pilot projects.
3. Reduced reliance on regular battery replacement for remote submerged devices.

Proposed Methodology

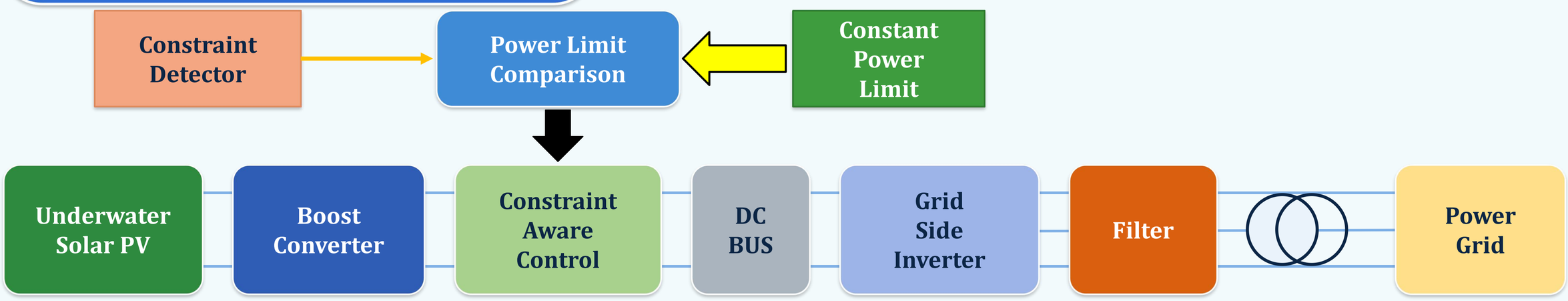


Fig-1.1: Proposed constraint aware control architecture for Grid-Connected Underwater Solar PV System

Conclusion

Underwater solar energy is a promising research area for independent low-power applications. Bangladesh's near-term potential lies in powering submerged monitoring, sensing and communication systems rather than producing large amounts of electricity. The proposed system combines underwater PV, battery storage and Constraint-Aware Energy Management to distribute power within battery SOC, load, power-limit, grid and power-balance constraints, improving efficiency and secure operation. Future experimental validation under Bangladeshi water conditions is essential.

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